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Psychopathologies of time: Defining mental illness in early 20th-century psychiatry

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Abstract

This article examines the role of time as a methodological tool and pathological focus of clinical psychiatry and psychology in the first half of the 20th century. Contextualizing 'psychopathologies of time' developed by practitioners in Europe and North America with reference to the temporal theories implicit in Freudian psychoanalysis and Henri Bergson's philosophy of *durée*, it illuminates how depression, schizophrenia, and other mental disorders such as obsessive-compulsive behaviours and aphasia were understood to be symptomatic of an altered or disturbed 'time-sense'. Drawing upon a model of temporal synthesis whereby in healthy individuals, a subjective temporal sense (*Ichzeit*, *durée*, or personal lifetime) was perceived and understood in relation to objective time frameworks (*Weltzeit*, clock-time, or quantitative time models like historical chronology), clinicians argued that mentally ill patients were unable to synthesize *Ichzeit* and *Weltzeit*, using variations in this disturbance to define specific pathological conditions.

Keywords

Henri Bergson, phenomenology, psychiatry, temporality, time

In 1908, Swiss-American psychiatrist Adolf Meyer declared that different physiological states were characterized by differences in 'time relationships' and 'should be expressed in terms of "psychological time"' (Meyer, 1912: 142). While psychologists had traditionally located the origins of mental experience in brain physiology, Meyer proposed

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the reverse. He asserted that experiences of 'psychological time' were the basis of physiological processes. Meyer's privileging of 'psychological time' as an essential mode of temporal experience implied that cognitive or mathematical models of time – such as clock-time, calendars, or historical chronology – were secondary to the psychic experience of temporality. Meyer was not alone in highlighting the importance of psychological time to lived experience. Starting in the 19th century, experimental psychologists had devised ways of testing human reaction speeds, the duration of perception, and conscious recognition of a tenth of a second (Canales, 2009). In the fin de siecle and the early decades of the 20th century, philosophers such as Henri Bergson, Edmund Husserl, J. M. E. McTaggart, and Martin Heidegger grappled with the nature of time, while novelists like Marcel Proust, James Joyce, and Virginia Woolf literally depicted the experience of temporality. In the same temporal milieu, Einstein formulated his special theory of relativity in 1905 and the general theory of relativity in 1915.

In the following three decades, psychiatrists and psychologists inspired by the findings of philosophy (particularly Bergson's work on duration) asserted the uniquely psychic nature of time consciousness. While Freud had emphasized the role of the past in psychoanalysis and French psychiatrists Pierre Janet and Charles Blondel had drawn attention to the importance of present-conceptions in mental life, a group of phenomenologically inclined psychiatrists during the interwar years began to focus on conceptions of the future. A key figure in this burgeoning field of time psychology was Eugène Minkowski (1885–1972), a Franco-Russian psychiatrist who practised in Paris. Inspired by Bergson's philosophy of duration, which underscored the incongruity between man's intuitive experience of time (as duration) and his tendency to spatialize time externally, cognitively representing it as a clock, calendar, or time 'line', Minkowski averred there was an irreparable discord between how man *described* time and how he *lived* time (*le temps vécu*). The Western tradition of representing time as a straight line as if its natural state took the form of a timeline or lifetime, Minkowski argued, produced an uncomfortable oscillation between the linear representation of temporality in discourse and the 'irrational' bodily experience of time, thus failing to account for the plethora of temporal phenomena that 'slipped' between these two aspects. Although historians have tended to focus on developments in timekeeping, time standardization and new transportation or communication technologies to understand the time of modernity (Galison, 2003; Ogle, 2015; Schivelbusch, 1986), Minkowski insisted that the problem of time was not intrinsically related to the mathematical or physical problems raised by the experience of simultaneity popularized by Einstein's theory of relativity (Minkowski, 1970[1933]: 17). Rather, he looked to contemporary philosophers like Bergson who were interested in the metaphysical nature of time to understand the experience of temporality as a cognitive process in the mind (that is, the psychology of time) as well as a means of situating oneself within the world (the phenomenology of becoming).

This article explores how the concept of future time became a central paradigm for defining and understanding mental illnesses in early 20th-century psychiatry. Following a brief discussion of time in the psychiatric work of Minkowski's precursors and Bergson's philosophy of *durée*, I present Minkowski's phenomenological approach to mental illness and trace how his time-based definitions of depression and schizophrenia influenced and were supported by concurrent research into time-based

disorders contrasting intuitive/subjective time (*Ichzeit*) and mathematical/objective time (*Weltzeit*) in Europe and America. I conclude that a 'futurist' approach to psychiatry was widespread in the decades before the Second World War, revealing how questions of time perception, the 'true nature' of time, and the metaphysics of the future significantly shaped scientific paradigms in early 20th-century psychology. The pathological experiences of time uncovered and developed by phenomenological psychiatrists not only constructed alternate ideas of past, present, and future; they also challenged normative conceptions of 'clock time' and suggested the artificiality or incompleteness of timekeeping systems in general.

Modernity and time

In classical psychology, psychic phenomena were believed to derive from physiological phenomena and accordingly were treated by physical methods (Thinès, 2007: 79). Until the turn of the century, psychology had been dominated by two fields: experimental psychology and physiological psychology, whose theories tended to reduce mental disturbances to physiology. Increasingly, however, this reductionist-empirical paradigm failed to account for the complexity of human perception and behaviour, particularly as the topic of inter-subjectivity became prominent in early 20th-century Germany through the work of Max Scheler, Edmund Husserl, Martin Buber, and Karl Löwith (Lanzoni, 2003: 160). This impetus gave rise to new schools of thought, notably psychoanalysis and Gestalt psychology, in the early 1900s. Historian Henri F. Ellenberger cites Ludwig Binswanger's paper 'On Phenomenology', given at the Société Suisse de Psychiatrie in November 1922, as the first sign of a tertiary approach: phenomenological psychiatry. Although trained in clinical psychiatry and subsequently in psychoanalysis, Binswanger suggested that Husserl's phenomenology could be applied to clinical psychiatry – the first step in his development of *Daseinanalyse*.

Nonetheless, the first paper on clinical phenomenology, 'Findings in a Case of Schizophrenic Depression', was published by Minkowski in 1923, describing a schizophrenic patient who announced daily that he would be executed in the evening. As Ellenberger points out, although the usual psychiatric approach would have assumed that the patient's time perception was distorted because of his delusions, Minkowski suggested the opposite. He claimed that his patient's delusions were manifestations of a 'more basic' temporal disorder in attitudes toward the future (Ellenberger, 1970: 842). Phenomenological psychiatry aimed to go beyond experimentally attainable curves or figures in order to arrive at the totality of human existence. Historically, psychology had deconstructed existence into 'pieces' of 'sensations, perceptions, ideas, sentiments and volitions' that were analysed separately, but phenomenological psychiatrists like Binswanger, Minkowski, and Karl Jaspers sought to understand the entire 'mosaic' of lived experience (Minkowski, 1926a: 554). These practitioners believed that applying philosophy to human psychology would enrich psychological knowledge (May, Angel and Ellenberger, 1958). In particular, they argued that phenomenology could elucidate the experience of time typically concealed to everyday life.

In her seminal account of the debate between Bergson and Einstein on 6 April 1922 regarding the essential nature of 'philosophical' or 'physical' time, Jimena Canales

persuasively illustrates how time was a critical topic of debate among early 20th-century intellectuals (Canales, 2015). Her work on the scientific debate surrounding the meaning and measurement of a tenth of a second, which Canales refers to as 'constitutive of modernity', further reveals how physics emerged as the 'privileged science of modernity' (Canales, 2009: ix, 181). Yet modernity was not only characterized by infinitely smaller time intervals or the pre-eminence of physics; it was also a period in which artists, scientists, science fiction writers, and playwrights grappled with the experience of time and the possibilities of its metaphysical existence (Ardoin, Gontarski and Mattison, 2013; Kern, 1983; Moses, 2014). Starting in the 1890s, Henri Bergson had highlighted the inadequacy of a purely scientific or quantitative model of time, conceptualized as the 'linear succession of temporal moments', for capturing temporality as it was experienced in daily life. Salvador Dalí's melting clocks in *The Persistance of Memory* (1931) or Italian experimentations with Futurist photodynamism (Bragaglia, 2008[1911]), as well as the temporal parallels or the intersection of past and present dramatized in plays like René-Henri Lenormand's *Le temps est un songe* (1929) or Virginia Woolf's novel *Orlando* (1928), all testify to a shared exploration of non-linear, subjective temporalities.

This obsession with time admittedly irritated some. In 1927, English art critic Wyndham Lewis condemned the popularity of the 'time-mind' and its manifestations in interwar literature, poetry, and painting, complaining how modernism's time proponents demanded the public 'see everything *sub specie temporis*' (Lewis, 1927: 5). Lewis argued that 'the "timelessness" of einsteinian [sic] physics, and the time-obsessed flux of Bergson, merge in each other, and . . . have conspired to produce, upon the innocent plane of popularization, a sort of mystical time-cult' (ibid.). He attributed the intellectual origins of the 'time-cult' to Henri Bergson's philosophy of *duréé* and Einstein's theory of relativity, then traced the dissemination of this obsession through works by James Joyce (*Ulysses*), Marcel Proust (*Remembrance of Things Past*), and Gertrude Stein, whom he condemned for attempting to locate temporal 'flux' in everything (ibid.: 8). Lewis regarded Bergson and Einstein's influence on the rise of the 'time-cult' as largely equivalent. Although he acknowledged Einstein's epistemological contributions, he nonetheless claimed that the esteemed physicist had formulated his theory of relativity within a culture already obsessed with time.

The 'time-sense' and temporal malfunction in early psychiatry: Freud, Blondel, Janet

It is perhaps unsurprising, then, that psychologists joined the temporal fray. The community of clinical psychologists and psychiatrists across Western Europe and North America who collaborated with Minkowski in this enterprise were uniquely positioned to uncover the 'true' nature of 'lived time' as psychic experience, responding to debates about the subjectivity of temporal experience that transcended early 20th-century physics and philosophy. Starting with Charles Blondel and Pierre Janet in the 1890s, clinical psychologists and psychiatrists had increasingly started to define, analyse, and interpret mental disorders based upon aberrations in supposedly 'normal' time experiences. These practitioners perceived pathological psychoses as particularly apt vehicles for exploring the 'intrinsic temporal structure' of mental disease, starting with memory. The pathology

of memory, which emerged as a distinct field of research in late 19th-century psychology through the work of Hughlings Jackson and Theodule Ribot, had precipitated a new-found focus on the 'dimension of time' alongside other means of differentiating memory (Danziger, 2008: 163; Ribot, 1881). As historian Kurt Danziger observes, both memory and heredity began to be conceived as 'layered structures extended in time, so that layers would be superimposed on earlier layers' whose relationship to the present differed: older memories were perceived as more 'stable' and 'fixed' compared to newer ones (Danziger, 2008: 162). More worrisome, time was increasingly viewed as 'reversible' – while it 'moved forward in evolution . . . it could also move backward in dissolution' (ibid.: 163). Researchers therefore began to explore the temporal phenomena of cognitive processes. In particular, psychiatrists developed Freudian theories of memory and Bergsonian notions of temporal subjectivity to argue that time was fundamentally a metaphysical problem of 'perception' and 'becoming' in the mind (Spiegelberg, 1972: 261; Straus, 1966). Research into pathological temporalities, it was argued, would not only shed light on the character, symptoms, and potential treatment of psychopathologies, but would also help elucidate the 'time-sense' (or *Zeitsinn*) typically 'concealed' to observations of normal life (Straus, 1947: 257).

Theories of temporal malfunction in psychiatric patients were indebted to Sigmund Freud's theory of past trauma, particularly his work on morbid memory disorders that suggested how the 'unit of time' for trauma patients was 'cut out' from a homogeneous perception of the senses, disrupting the typical relationship between permanent traces (*Dauerspuren*) and memory (*Erinerrungsreste*) (Armstrong, 2005; Freud, 1971[1917]). The Freudian canon postulated that individuals suffering from shell shock and other psychic maladies experienced a warped temporality in which traumatic illness was the result of a disruption in the cerebrum's protective barrier. This theory, in addition to the wartime experiences of soldiers starting in 1914, renewed the focus on memory and an individual's time-sense in American, British, French, and German clinical research (Straus, 1947: 259). Freud's notion that memory simultaneously 'constituted' and 'undid' the individual, forming a sense of the self alongside the capacity to splinter experience in the form of mental disturbances that interrupted or fractured conscious temporality (Ffytche, 2012: 240; Forrester, 1980; Terdiman, 1993: 240), constructed a theory of pathological time based upon orientations toward the past. Yet while Freud focused on past concepts in the temporal pathologies of psychoanalysis (trauma and repression), phenomenological psychiatrists theorized that patients were ill because their temporal relation to the future was diseased.

Although historians have published widely on the function of memory in relation to regimes of historicity and memorial culture in Europe as a result of the two world wars (Assmann, 1999; Bann, 1995; Gross, 1992; Hartog, 2003), these studies tend to situate memory within a culture of time implicitly tied to a model of modernity characterized by the experience of 'trauma' or 'shock' from a sociopolitical perspective, and thus an 'anxiety-producing entity that must be thought in relation to management, regulation, storage, and representation' (Armstrong, 2003; Doane, 1996: 314). My analysis does not engage directly with memory in the fin de siecle, which is a vast and well-established topic in its own right (Le Goff, 1992; Nora, 1984–92; Terdiman, 1993). Instead, I am concerned with how studies and theories of future time perception in early psychology fit

into broader explorations of temporality during this period. The starting point for phenomenological psychiatry during this period derived from Freud's assumption that 'normal' and 'abnormal' (or 'diseased') psyches existed, with the associated belief that psychological disorders were the product of some sort of disturbance. For Minkowski and likeminded researchers, mental illnesses were caused by an inability to conceive of an open future within the external framework of world or clock-time. As such, conceptions of past and present were less integral to the origins of psychopathologies. Indeed, despite his manifest admiration for Bergson, Minkowski largely ignored the philosopher's work on *Matter and Memory* (1896).

A second approach to time pathologies focused on conceptions of the present. Psychologists had already observed that 'consciousness of change' lay at the root of man's experience of time and that awareness of a change in 'order, simultaneity, and succession' was acquired through accumulated experience and reflection (Leighton, 1908: 561). Although it was possible to conceive of other organisms having 'minuter or coarser' time-perceptions than man, the ability to orient one's personal time consciousness within the tempo of the wider world, and thus the ability to categorize one's temporal experiences in terms of the three dimensions of past, present, and future, was an implicit requirement of psychological normality (Harrison, 1934; Nilsson, 1920). This 'act of synthesis' required the active engagement of consciousness insofar as a subject was aware of the passing present in cognizance of the wider present subsequently conceived (Bromberg and Schilder, 1936; Schilder, 1935). Psychiatrists theorized that the actually experienced present was a moment of becoming with 'no sharp delimitation before and after' that was only retroactively conceived (Leighton, 1908: 561–2), echoing Husserl's philosophy of the 'living-present' with its thickened retentions (pastness) and protentions (futurity) (Sokolowski, 2000). The lived present contained an inherent breadth stretching from memories of the remote past, to the immediate past, to anticipations of the future. A 'normal' memory was thus defined as the ability to conceive of previous times or experiences within the contextual horizons of the present without 'losing' oneself or living entirely in the past, à la Freud. Hence Adolf Stöhr defined the *Zeitproblem* as a 'borderland' or 'border sensation' in the psyche between past and future (Stöhr, 1910).

Pierre Janet's *L'évolution de la mémoire et de la notion du temps* (1928) was the first psychological monograph devoted to time in psychopathology, with particular reference to experiences of the present. Charles Blondel's earlier book, *La conscience morbide* (1914), had covered a handful of case studies relating mental maladies to time, but only a few chapters were devoted to 'morbid' temporalities, and Blondel had not generally defined mental illness in terms of time. His book was nonetheless inspirational for Janet, who repeatedly referenced him. Janet argued that a concept of time did not develop from physical interactions with space but from secondary behaviours related to affective life that derived from a psychology of effort (starting, persevering, achieving) grafted onto memory and recall (Janet, 1928: 24). Time was therefore a 'psycho-biological' structure related to belief, which could be philosophical or historical in nature, and based on concepts like progress, evolution, and creation (on this second point Janet was indebted to the phenomenology of Jean-Marie Guyau) (Guyau, 1890; Minkowski, 1950).

Janet began with the question of how man conceived of time as a real entity when he constructed a 'before', an 'after', and all the times 'in between' in the mind and consequently through language (Janet, 1928: 306). He suggested that this was achieved partly by the equation of the 'present' with 'existence' in childhood. Contrasting 'healthy' and 'unhealthy' experiences of the present, Janet concluded some psychiatric patients were ill because they were convinced that the present was unreal. Patients who suffered from temporal abnormalities like memory aphasias nevertheless insidiously employed the word 'present' to seem like everyone else, when 'in reality, they [did] not know how to construct it, and [did] not', because they did not feel the present as an action that would end or be completed (ibid.: 334). Janet used various examples of aphasic and hysteric patients to support his claim that psychoses were related to abnormal orientations toward the present. Janet hypothesized that there was no difference between 'present' and 'future' in the mind of his schizophrenic patient Madeleine, for example, because she could not speak without prophesying the destruction of Paris by fire, water, and destruction at the feet of red horses, yet was unable to say exactly when the French capital's demise would arrive. Conversely, patients suffering from melancholia were incapable of applying proper dating systems, like days according to the Gregorian calendar or annual celebrations such as Christmas, instead inhabiting 'fictive presents' that they created in the past and returned to via recollection (ibid.: 338). Janet concluded the dimensions of past, present and future were 'blurred' in mental illness, losing all value to the extent that present time 'no longer exist[ed]' (ibid.: 339). This temporal elision gave birth to 'delusions'. By understanding how patients 'located' or confabulated memories in the present, Janet argued, clinicians could understand the temporality of their patients' ideas and thus shed light on the aetiology of their mental illnesses (ibid.: 25–6).

Bergson and time

A philosopher colleague of Janet's at both the Sorbonne and the Collège de France, Henri Bergson also showed an interest in memory and time. Starting with *Essai sur les données immédiates de la conscience* (1889; trans. *Time and Free Will*, 1910), Bergson proposed that the inner self was accessed only by intuition and that the first and primary sense of intuition was becoming in time, which he called 'duration'. Unlike Kant, who suggested that time was an a priori form, Bergson proposed that the 'true' structure of time was revealed only in consciousness. Breaking with classical philosophy and 19th-century positivism, he argued that time had been spatialized in 19th-century modernity via mathematical or mechanical models of time (such as clocks) that failed to account for time as lived experience (Minkowski's *temps vécu*). Separate to the objective 'reality' of Newtonian or mathematical time, which he viewed as an intellectual construct, Bergson argued that psychological time existed prior to cognition as an intuition or sensation of duration (*la durée réelle*). He defined duration as the pure heterogeneity or 'qualitative multiplicity' of mental states that succeed each other without distinction (Bergson, 1910: 226). This qualitative interior flux, or *temporality*, was subsequently differentiated and expanded through abstract thought into spatial concepts and linguistic metaphors that

objectified time, eventually rendering it a quantitative entity in the physical world (Bergson, 1910; Muldoon, 1991: 254–5).

Although Bergson did not provide a distinct 'logic of time', he nonetheless proposed that past, present, and future were not empirical realities, but different psychological 'states' that remained distinct even when conceived simultaneously (Vetö, 2005: 6). Rather than faithfully reproducing the past or succeeding the present moment of perception, memory was contemporaneous to perception and constructed in the present (ibid.: 13). Although duration was the essential and original experience of time, Bergson acknowledged that modern life required the ability to synthesize internal duration with external clock-time (Bergson, 1911: 188). It is this synthesis, in addition to Bergson's emphasis on the psychological origins of the time-sense and the cognitive creation of time orders, that gave shape to psychiatric definitions of time-based pathologies in the early 20th century.

Minkowski praised Bergson for applying philosophical methods to psychology in order to better understand lived experience. The function of psychopathology, he suggested, was to verify whether Bergson's ideas could shed new light on psychological problems that current methods failed to understand (Minkowski, 1926a: 555–6). Phenomenological psychiatrists predicated their analyses on the philosophical premise that all human experience intrinsically contained a temporality. Minkowski and like-minded phenomenological psychiatrists suggested that individuals carried two parallel time-senses: an internal sense of time or temporal subjectivity akin to Bergsonian *durée* that included Freud's theory of memory (*Ichzeit*), which in healthy individuals operated parallel to, and was rendered consciously sensible in relation to, world-time (*Weltzeit*) as measured and maintained by chronometers and historical chronology (Klien, 1917). This resulted in the subjectivity of time experiences (such as periods seeming longer when bored) as well as an individual's ability to narrate past experiences, conceiving of events taking place in a particular life-cycle and situating that life-cycle within the broader chronologies of human history or cosmology. Combined, these produced a 'wholeness' of the state of time, implying that a 'healthy' time-sense included the ability to synthesize *durée* within the framework of mathematical time that 'ran' the world (ibid.: 309; Moskalewicz, 2018: 5; Straus, 1928).

Like Bergson, psychologists claimed that individual orientation within clock-time was necessary to exist in the world even if it was secondary to duration in immediate experience. Psychiatrist Ulrich Sonnemann explained this halfway stance as a sensitivity to the sensory world external to the body, where clock-time was crucial for social interaction, combined with the 'psychosomatic conditions' of the individual – what Nathan Israeli called a 'syntonic' reaction of simultaneous social and personal temporal processing (Sonnemann, 1954: 295). Minkowski thus averred that man experienced time in life through both a *stable* (clock-time) and a *dynamic* (duration) aspect (Minkowski, 1970[1933]: xxiv). Although the hegemony of clock-time was 'one-sided', clock-time was still a necessary tool for prognostication. Psychiatrists concluded that temporal perception was a twofold process, both part of the 'outer world' and 'an experience within ourselves' (Gillespie, 1936: 515).

Nevertheless, by asserting that time was subjectively experienced – one had to mentally synthesize personal duration with clock-time – and objectively synthesized with

world-time, psychiatrists claimed an epistemological authority that challenged physical and philosophical definitions of temporality. By defining time as an inherently psychic function, they were able to classify 'anomalous' time behaviours that were erroneously 'experienced' or 'invented', as British educational psychologist Mary Sturt wrote about states of dreaming or ennui (Sturt, 1925: 10–11). Unlike Freud's emphasis on the past in psychoanalysis or Janet's research into pathologies of the present, Bergson's philosophy of *durée* offered early 20th-century psychiatrists a new model for understanding mental disorders focused on the self 'becoming' within a flow of time that was oriented toward the future. Assuming that subjective experiences of time existed in parallel to increasingly global clock-time, psychiatrists suggested that perceptions of time were produced through a dynamic process in which *Ichzeit* was continuously synthesized with conventional models of *Weltzeit*. In so doing, they defined the parameters of a 'normal' time-sense and produced a discourse of pathological temporalities based on orientations toward the future that continued to shape definitions of mental illness in clinical psychology through the 1940s and 1950s (Straus, 1928, 1947: 258).

Eugene Minkowski and psychopathologies of time

Minkowski's pioneering contribution to the field, *Le temps vécu: Études phénoménologiques et psychopathologiques*, translated into English as *Lived Time* (Minkowski, 1970[1933]), was a miscegenation of Bergsonian *durée* with developmental psychology that claimed emotions were the organizational principle of temporal experience (De la Harpe, 1939: 142; Minkowski, 1926a, 1926b, 1927). Although unorthodox today, Minkowski's recourse to phenomenology in his clinical practice and psychiatric theory reveals the extent to which phenomenology shaped early 20th-century psychiatry – as can also be found in the work of Ludwig Binswanger and Karl Jaspers (Andreason, 2007; Schmitt, 1986; Spiegelberg, 1972). Minkowski emphasized a problem-centred psychiatry wherein phenomenology was applied as an 'intermediary' between psychophysics and pure phenomenology (Tatossian, 2007: 18). His approach had a profound influence on modern French psychiatry. Between 1920 and 1970, continental psychiatry was significantly more phenomenological than its Anglophone counterpart, and Minkowski's ideas were widely studied in France. He co-founded the journal *Évolution Psychiatrique* in 1929, and his scholarly output influenced later 20th-century psychiatrists like Arthur Tatossian, Rollo May, R. D. Laing, and Jaques Lacan. The significance of Minkowski's ideas was recognized beyond psychiatry; in 1949, Merleau-Ponty praised Minkowski's pioneering introduction of phenomenology to existential analysis in France (Sass, 2001: 253). Yet Minkowski's influence on time-based models in European psychiatry, as well as the widespread nature of temporal approaches in early 20th-century psychology, has been little studied, even though the temporal nature of pathologies like depression has returned to scholarly debate (Abe and Raballo, 2013). The impact of Bergsonian philosophy and Minkowski's adaptation of Bergson's theory of time to clinical practice therefore remains a compelling presence in psychiatry today (Fuchs and Van Duppen, 2017).

Minkowski played a central role in shaping how subsequent researchers conceptualized and defined depression and schizophrenia according to supposedly 'abnormal' time

perception based on a binary model of *Ichzeit* and *Weltzeit*. Earlier work by Swiss psychiatrist Paul Eugen Bleuler and Russian-American psychiatrist Aaron Joshua Rosanoff had explored time in psychopathology, but Minkowski made his career as a specialist in the sub-field of time pathologies. He departed from Janet's focus on 'present', 'past', and 'future' disorders by defining mental diseases according to specific distortions (*le trouble générateur*) in temporal orientation toward the future. Prior to the First World War, Minkowski's papers were written, and as he noted, thought through, in German. From 1918 he lived and worked in France, claiming to think exclusively in French. Yet Minkowski's works were cited in English research publications shortly after their debut, indicating how quickly his ideas entered British and American research. *Lived Time* consequently inspired a series of research programmes by German, French, American, and British psychologists that explored the pathological time-orientations of their patients.

Minkowski began with the philosophical premise that the temporal dimensions of past, present, and future in mental life were organized through phenomenological categories such as spatiality, materiality, and causality, borrowing Henri Bergson's argument in *Time and Free Will* that the most fundamental aspects of experience were non-rationalistic and therefore non-quantifiable, a premise that opened him to criticism from experimental psychologists who preferred the hard facts of statistics (Israeli, 1932: 486; Minkowski, 1970[1933]: xix–xx). Drawing upon his clinical work at Burghölzli University Clinic in Zurich and in Parisian hospitals, the second half of the book used specific case studies to analyse the spatiotemporal structure of mental disorders. Minkowski defined mental illnesses including depression, schizophrenia, and obsessive-compulsive disorders based on a dualistic framework of qualitative duration (*Ichzeit*) and quantitative clock-time (*Weltzeit*). He concluded that psychoses were the product of a breakdown in the ongoing synthesis of *Ichzeit* with *Weltzeit*, resulting in a loss of future-orientation that prompted patients to compensate with temporal delusions (Minkowski, 1970[1933]: 24).

For Minkowski, healthy time relations were produced through a conception of the present as an extended 'now' stretching out in duration with limits that were 'fluid' and 'supple' and oriented toward the future. The movement from the 'now' to the 'present', which he called the 'homogenization' of time, differed from individual to individual, although in healthy psyches this was synchronized with clock-time. This dynamic temporality or temporal unfolding constituted the essence of being. In other words, the experience of time as a constant flow of becoming (or *devenir*) oriented toward an open future constituted healthy existence (Urfer, 2001: 280). Healthy psyches were able to recognize their 'present' in relation to a 'past' and 'future' while being able to reconcile these personal time dimensions with social schemas of clocks, calendars, or historical chronologies. Symbolic representations of time like clocks and calendars enabled man to isolate this temporal flow and reflect upon specific experiences (ibid.). Healthy individuals were able to continuously and supplely synthesize personal/affective time and public/objective time by accepting and accommodating this flow of life, which Minkowski called *syntony*. By contrast, schizoid personalities were unable to resonate with their spatiotemporal surroundings and, through a 'loss of vital contact with reality', moved independently out of sync with society and time (Minkowski, 1970[1933]: 273–5).

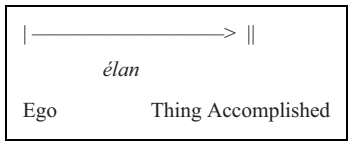

Figure 1. Minkowski's rendering of 'personal *élan*' (Minkowski, 1970[1933]: 46).

While an excess in either direction would lead to psychiatric disorders (manic disorders in cases of an excess of syntony, schizophrenia or depression in a schizoid temporality), the dynamic tension or movement between the two was essential to life. Adapting to social contexts and to one's surroundings was equally as important as the pursuit of independent goals and creative projects that enabled self-realization (Minkowski, 1927: 162).

Central to self-realization and to Minkowski's theory of lived time was temporal orientation toward an anticipatory future or imminence that he developed through Bergson's concept of *élan vital*, a 'vital force' in consciousness related to intuition and the flow of inner time. It was 'through the future that the Self affirm[ed] itself as a living being', he asserted (1933), by experiencing personal becoming within the flow of time. Minkowski refined Bergson's concept to fit his psychological focus as 'personal *élan*'.

According to Minkowski, *élan* functioned as the future-oriented bond between an individual and his or her becoming within the environment of experience (Urfer, 2001: 280). The bond was future-oriented insofar as the resonance between the individual and his or her becoming was characterized by imminence or open anticipation of what was to come. This dynamic directed human activities toward the future and toward self-realization (Van Duppen, 2017: 391). The concept of personal *élan* was highly metaphysical (see Figure 1); Minkowski variously defined it as 'universal destiny' or the potential for 'spiritual communion' with a transcendental entity that also guided man (Minkowski, 1970[1933]: 51). This 'futurism' was heavily tinged with social utopianism and cooperation, particularly in the 'sixth aspect' of future orientation, which Minkowski explained as 'the quest for ethical action', enabling the individual to construct a 'contact with reality' or 'lived synchronism' that generated a feeling of moving in time with one's environment and personal rhythm known as 'ambient becoming' (ibid.: 69–70). Here the contrast between syntony – the principle by which man could 'vibrate in unison with the environment' through reconciling different temporalities (such as recollecting memories in the present, anticipating events in the future, and experiencing the temporality of consciousness alongside and within the temporality of clock-time) – in the normal psyche, and a schizoid temporality that was unable to harmonize personal *élan* with external reality, laid the grounds for a medical discourse of the pathology of time.

Pathological temporalities, by contrast, were unable to shift seamlessly between synchronization and self-realization or conceive of a personal *élan*. Instead, they manifested an inability to seamlessly reconcile the interface between temporal modes (*Ichzeit* and *Weltzeit*) or 'lived' excessively in one of the three temporal dimensions of 'past', 'present', or 'future', thus standing still instead of flowing in time. The origins of psychopathologies, Minkowski concluded, lay in a spatiotemporal *trouble générateur*

specific to every psychiatric disorder. While healthy individuals experienced personal becoming (*devenir*) as a sense of moving toward the future (Van Duppen, 2017: 390), mentally ill patients exhibited a 'loss of vital contact with reality' that affected their spatiotemporal experience by impeding the coherence of being-in-the-world (ibid.: 386). Lacking this contact with reality, the individual's orientation toward the future changed, thereby altering the key structuring dimension of human existence (Urfer, 2001: 282).

Minkowki's phenomenological approach drew the attention of influential psychiatrists, including Ludwig Binswanger, Karl Jaspers, Franz Fischer, Erwin Straus, Aubrey Lewis, and Viktor von Gebsattel, who were part of the bid to create a new, phenomenological psychiatric tradition based on the intersubjective encounter between patient and doctor. As Binswanger observed, psychoanalysis problematized psychiatry by revealing how the latter elided the metaphysical distinction between body and soul (Freud and Binswanger, 2003: 172; Lepoutre, 2014: 109). Like psychoanalysis, phenomenological psychiatry presented a holistic science that combined the mental and the physical, yet suggested that understanding structures of being-in-the-world were more important than consciousness or the unconscious for comprehending mental life (ibid.: 113; Stierlin, 1963). Time, in particular, was isolated as an essential dimension of existence that could be studied in the clinic or hospital (Schumann, 1898; Volkelt, 1925). As Binswanger observed, temporality was the 'fundamental horizon of all existential application'; von Gebsattel argued that psychiatrists ought to uncover 'lived time' (*temps vécu* or *gelebte Zeit*), rather than the experience or consciousness of time (Von Gebsattel, 1939). In other words, he suggested (and Binswanger agreed) that they should attempt to understand time as it 'happened' (*Zeitgeschehen*) to a person internally versus time as it was 'heeded' cognitively by the individual (Binswanger, 2004[1958]: 301–2). As a consequence, phenomenology, and in particular the phenomenology of time, became an increasingly influential tool in the development of modern psychiatry in Central Europe, Britain, and America (Herzog, 1992; Kusch, 1995; Spiegelberg, 1972). Concurrently, definitions of personality based on the self's temporal orientation were advanced in characterology (Bouman and Grünbaum, 1929; Lanzoni, 2003: 161–2).[1]

Alongside Binswanger and Minkowski, a third psychiatrist sympathetic to the programme of studying psychological time was Erwin Straus. Straus taught at Berlin's Humboldt University (1931–5) before emigrating to America in 1938, where he taught at Black Mountain College and then worked as Director of Clinical Psychiatry at the US Veterans Administration Hospital in Lexington, Kentucky, from 1946 to 1961. Like Binswanger, he was interested in the epistemological foundations of psychoanalysis, but drew his own analyses from Husserl's phenomenological discussion of man's life-world. Straus believed that individual time experiences differed vastly across society. He suggested that clock-time could not be the primary experience of time, since impressions did not follow each other consecutively. Though moments were important, they were merely 'phases' within 'a larger whole' that was always 'incomplete' and always attached to adjacent moments in both the past and the future. These holistic moments represented 'the whole of our vital, social, historical existence', confirming Bergson's evaluation that 'normal' life, and therefore a normal temporal order, was a 'state of becoming' or duration (Moss, 1979; Piéron, 1923; Von Gebsattel, 1954a: 128–44). Marcin Moskalewicz has written extensively on Straus's 'unified view of time'.

Moskalewicz argues that Straus struggled with the tension inherent in the dualistic model of time essential to the phenomenological definition of lived experience as an unstable equilibrium between *Ichzeit* and *Weltzeit*. Straus eventually concluded that personal/experienced time and clock-time were two halves of the same whole, unified through his notion of 'today' – much the same way that Minkowski defined 'normal' temporality as the ability to synthesize *Ichzeit* and *Weltzeit* simultaneously. For Straus, however, personal time was not purely psychological; it was also biological or ontological, a primary condition of human experience that Straus came to view as a necessary subordinate to clock-time (Moskalewicz, 2018: 6–9). Although he stressed that a symptom of psychotic states was an unhealthy focus on the past, Straus never distinguished mental disorders based on specific temporal experiences (ibid.: 11).

German psychiatrist Franz Fischer advanced a similar schema of time-relations (*Zeitbeziehung*) where the mind's commingling of *Ichzeit* with *Weltzeit* produced an awakening that allowed an individual to move through the world, borrowing philosopher Max Scheler's notion of *Selbstbewegungsmacht*, or autonomy through self-progress (Fischer, 1929). Minkowski, Straus, Fischer, and Aubrey Lewis of the Institute of Psychiatry in London suggested that when *Ichzeit* and *Weltzeit* were in harmony, normal life proceeded. However, the two temporal worlds did not always need to be commensurate or symmetrical: it was possible for ego-time to proceed more rapidly than world-time and vice versa (when one was unhappy, bored, or morose, for instance, clock-time appeared to pass more slowly). Uncovering the 'psychology of time' was crucial to understanding how mental processes allowed subjects to recognize the present, judge the passage of time, and relate time to 'other contents of consciousness' (Lewis, 1932: 612). These authors urged psychiatrists to seek 'the spatiotemporal substratum of the whole psychopathological syndrome', not only in order to define and understand mental disorders based within an intellectual framework of abnormal temporalities, but also as a means of understanding the self-actualization of man in general (Minkowski, 1970[1933]: 252; Sonnemann, 1954: 295). The following three sections accordingly detail the temporal features of depression, schizophrenia, and other mental disorders, revealing how conceptions of future time were utilized to rationalize and diagnose psychopathologies.

The temporality of depression

In 1914, Blondel had observed that people suffering from mental afflictions often had very different conceptions of time compared to society at large, exhibiting the belief that there were no 'periods', nor any such things as past, present, or future. Instead, patients claimed that 'Time . . . [was] a tunnel indefinitely prolonged, an interminable underground that present[ed] no exit, no way out, no variety' (Blondel, 1914: 45, 216). Blondel used the case study of a patient named Charles, whom he treated from April to November 1910, to understand the time-sense in depression. While Charles recognized the date, Blondel anxiously noted, the *idea* of a date remained vague and fluid for him. Instead, Charles appeared to live 'day by day . . . like an animal' in a time that he professed to feel 'immense, interminable'. Charles had 'lost the notion of time: night, day, yesterday, today, tomorrow' were endlessly identical (ibid.: 44). Blondel concluded that a classic symptom of depression was a loss of the 'typical' notion of time. Unable to

orient time orders or events in a conceptual continuity (a timeline of one's life, for example), they were enveloped by an understanding of time as an infinitely undifferentiated mass (ibid.). Here Blondel made an intriguing hypothesis that later authors failed to pick up. Noting how depressive patients were unable to synchronize the 'beats of a pendulum' in a series or to conceive of individual progress in the form of thoughts proceeding and evolving in time, he suggested that 'sick' patients had returned to an 'immeasurable' (non-serial) form of internal *durée* that had been lost to those with 'regular' time-conceptions – that is, those who were able to reconcile duration with clock-time. Unlike Minkowski, Janet, and others who succeeded him, Blondel therefore bestowed unhealthy temporalities with the unique ability to live psychological or subjective time in an evolutionarily authentic, less mechanistic form. It was for this reason that depressed patients had lost the character of *social* time: the time of religious ceremonies, eras, or dates that 'healthy' psyches applied in a quotidian fashion (ibid.: 45).

Pierre Janet accepted Blondel's conclusion that patients suffering from depression were unable to conceptualize clock-time, but queried why they had lost this ability (Janet, 1928: 340). Minkowski resolved Janet's dilemma using the dichotomy of 'immanent' and 'transitive' time, borrowed from contemporaneous work by Straus and von Gebsattel, which he refined as *Ichzeit* versus *Weltzeit.* Minkowski claimed that a discrepancy between these two modes was most extreme in pathological states of 'endogenous depression' wherein *Ichzeit* slowed down to an extreme rate, which Straus attributed to biological 'inhibition' growing more acute as it slowed the flux of internal time and intensified the power of the past (Minkowski, 1970[1933]: 298; Straus, 1928). Minkowski explained this in purely psychological terms: the pathological deceleration of *Ichzeit* in depression produced a temporal state wherein the patient's normally future-oriented perception focused overly on the past. Hence the feeling of temporal stasis cited by patients suffering from depression who claimed to feel 'displaced' in life (Minkowski, 1970[1933]: 298). While they could sense time passing, they lacked 'the sensation of following the movement' (ibid.: 332). One of his patients testified:

> I exist in the present only in idea but neither in feeling nor emotively. I am obsessed by the past. The images of my past go by like the scenes in a cinema, but I do not attach them to the present; I observe them like a spectator. . . . Often there is in me a kind of rolling by of past events. I experience past events completely, as if they were present. It is a half-dream of the past. . . . A constant progression of the past occurs in me. (ibid.: 335)

Minkowski concluded that his patient felt himself to be 'sick in relation to time', and observed that his symptoms related to a malfunction in lived duration, an inability to live 'contemporaneously' in present events or look to the future. The result was a dissociation between past, present, and future, as well as a 'deformation' of memory that aggressively imposed itself on the present to the point where it almost 'usurp[ed]' the latter (ibid.: 337; Schilder, 1935: 265). He and other psychiatrists emphasized how depression enfeebled a patient's orientation toward everyday temporality:

> Is there a future? Before I had a future, but now it shrinks farther and farther away. The past is so pushy, it throws itself on me, pulls me back. I'm like a machine that stands on the spot

> and works. It is labouring, [so] that almost everything snaps, but it stands still . . . by which I mean there is no future and that I will be thrown back. (Hannibal, 1955: 609)

Minkowski accordingly explained the temporality of depression almost as a form of ego-time-dissociation. If the category of the 'present' or 'now' represented a moment of becoming specific to every individual that was contextualized within an abstract order of time (that is, historical chronology or clock-time), personal existence could be situated within the broader rhythms of society, history, or cosmology.

Unlike calendric or cosmic time, personal time was not homogeneous, resulting in a 'deep contrast' between at least two modes of temporality that was psychologically experienced as 'accordance' or 'discordance'. Time passed 'quickly' or 'slowly' according to the subjective evaluation of the individual, as psychologists had repeatedly observed (Straus, 1947: 256). In depression, however, the 'basic structure of space and time, or better of the world in its spatial and temporal aspects, [was] altered' such that a patient's relationship to the world (*Weltzeit*) was irrevocably defective: patients felt both internal and external time as entirely static. Unable to see the past or distinguish it from the present or the future, patients claimed they could not 'think of past and the future; it all seems a blank' (Lewis, 1932: 616). Another melancholic patient similarly explained that 'the whole day seems exactly the same. If I look at the clock it doesn't mean anything to me at all' (ibid.: 611). Clinicians concluded that the synthesis or 'reintegration' of personal time (*Ichzeit*) and clock-time (*Weltzeit*) seemed to have failed in depressed patients (Straus, 1947: 257). Unable to experience the present as a continuation of the past 'gnawing' into the present as Bergson had suggested, the depressive's temporality 'lagged' behind *Weltzeit* (ibid.). This amounted to what Otto Hannibal described as a 'bloating' or 'standstill' of space and time (Hannibal, 1955: 610; Stewart, 1911: 8).

Melancholics thus appeared to suffer from a 'pathology of becoming'. All three orders of past, present, and future were affected ('diseased', in the eyes of their medical handlers) in depressed patients. To the latter, the present seemed remotely past, while the past itself appeared not to move. More troublesome, the future appeared too remote to be conceivable; it was not 'seen' by patients who were consistently past-oriented or who rejected time (Janet, 1928: 333). Speaking the language of Bergsonian duration, psychiatrists concluded that the flow of time became 'homogeneous', lacking any 'break' or differentiation for depressed patients, who experienced neither 'beginning' nor 'end'. Subjectively, depressives felt only an overwhelming past, and even more worrisome, some appeared to distrust the measurement of objective time, observing clocks as deformed mechanisms or claiming that clock hands moved backwards (Straus, 1947: 255–6). Phenomenological psychologists increasingly came to define depression as a disease wherein the specious present was infinitely prolonged to the point of appearing static, hence the over-estimation of time intervals by depressed patients, who extended a subjective sense of the mental present (or *psychische Präsenzzeit*) (Meerloo, 1935: 234). This deceleration of *Ichzeit* was accompanied by an inability to reconcile lived time with clock-time such that patients could conceive of neither past nor future.

The temporality of schizophrenia

Unlike depression, which was characterized by an intensification of 'pastness' and the inability to conceive of an open future, psychiatrists defined schizophrenia as a 'loss of vital contact with reality' expressed through (a) feelings of immobility and (b) 'morbid rationalism', an obsessive reflection on mathematical time (*Weltzeit*) due to the absence or loss of fluid becoming (*Ichzeit*), frequently manifested as a pathological disbelief in time itself. Like depressives, schizophrenics experienced a deficiency in 'lived time' and were unable to contemplate possible futures. For the latter, however, this pathology was often precipitated by a sense of a hyper-accelerating present that prompted schizophrenics to doubt the very concept of time itself. Patients believed that time was 'meaningless', an orientation that Fischer called *sinnlos*, and accused staff of manipulating timekeeping practices in the clinic or sanatorium, a manifestation of their recourse to a hyper-mathematized time (Fischer, 1929: 253; Minkowski, 1970[1933]: 186). Psychiatrists observed that these patients frequently expressed a desire for stasis:

> I love immutable objects, things which are always there and which never change.... The past is the precipice. The future is the mountain. Thus I conceived of the idea of putting a buffer day between the past and the future. Throughout this day I will try to do nothing at all. I will go for forty-eight hours without urinating. (Minkowski, 1970[1933]: 279)

Fischer noted that his schizophrenic patients, perhaps seeking to rationalize the temporal immutability so desired by Minkowski's subject above, constantly doubted the reality and meaning of clock-time (Fischer, 1929). One patient declared the clock had 'nothing' to say to him and asked how he should relate to it when he felt like time was going backwards, repeating itself again and again (Minkowski, 1970[1933]: 284). Indeed, a sense of recurrence was echoed by many schizophrenics, who claimed to 'live in eternity' where time no longer flowed. Unable to imagine past, present, or future, they felt themselves and time to be 'immobile' (ibid.: 285–6; Fischer, 1929: 545):

> Everything is like a clock. Time unfolds like a clock; the spectacle of life is like a clock. My life is now divided like the clock and time. I haven't anything more to say. It goes tick-tick morning, noon, and night, past, present, and future. It keeps happening again. There is always a little life, and then all that comes back. (Schilder, 1935: 263)

A perception of time's immobility was related to expressions of morbid rationalism. One of Lewis' patients, who complained of an inability to estimate time, challenged the standards of temporal normality implicitly represented and explicitly enforced by his psychiatrist. His paranoia about standardized time in the sanatorium is vivid:

> I can't say what time it is because it's an artificial day; what you call a day with the artificial day is very much shorter than the ordinary day. The time goes very much quicker.... I noticed my watch was accelerated.... What I mean is this; since we had breakfast this morning, according to your time it is eight hours, isn't it? Well, we haven't had eight hours since this morning by Greenwich time. The time you keep here isn't Greenwich time. Yours is only a quarter of the real time.... Probably in my months it'd be a couple of months since

> I came here, in what I call the ordinary time. But, of course, in your reckoning it'd be eight months, what you call a month. . . . Time in the sense of being heavy on your hands is terrible here, I can't do anything. By Greenwich time it goes very quick. But (considering) whether you find the moments interesting, time passes very slowly: every moment that passes is, you know, tedious and wearisome. Time in the sense of a period, though, is very quick. It would be about August, 1930, now by Greenwich time. I'm certain because I know Greenwich time couldn't have gone as quickly as July, 1931 [the date of the interview]. . . . Figuratively speaking it seems years since I was out in the normal world. . . . I never know any moment what is going to happen. It's the most terrible outlook I've ever had to look to. It's all perpetual. I've got to suffer perpetually. (Lewis, 1932: 617–18)

The testimony of Lewis' patient is striking for two reasons. His remarks challenge the hegemony of objective timekeeping systems like Greenwich Mean Time (GMT) while revealing the hyper-reflective mathematization of time exhibited by schizophrenics, conveyed by his paranoia about time-reckoning systems (the 'accelerating' watch or his contention that the clinic does not operate according to GMT). Although both depressives and schizophrenics were unable to synchronize *Ichzeit* and *Weltzeit*, Minkowski theorized that the immobility and hyper-rationalization of time manifested by schizophrenics derived from their inability to sense or feel a vital connection with ambient becoming. Having lost touch with personal *élan* and thus a sense of being-in-the-world (or duration), schizophrenics remained fixated on trying to measure and control time (Urfer, 2001: 283). While in daily life the measurement and observance of time was a normal behavioural tool, in pathology this morbid rationalism overwhelmed all attempts to spatialize time.

Schizophrenic patients at other clinics likewise resisted clock-time, accusing staff of 'playing with the clocks' to the point that some patients claimed whole days were 'missed' in a row, believing that elections held two days before their psychiatric assessments had actually taken place two or three weeks prior (Israeli, 1936: 57–8, 62; Lewis, 1932: 618–19). Fischer discovered an identical disbelief among his patients, who wondered if the orderlies were playing a trick on them and 'daemonically' turning the time back. Some questioned the 'ordinariness' of ordinary time: did orderlies 'still have ordinary time?' they asked (Fischer, 1930). Janet, too, discovered his schizophrenic patients harboured grave suspicions about clock-time estimations in the psychiatric ward. He likewise concluded that they suffered from an unreality of time, experiencing the present far more quickly than normal psyches – the opposite of, say, the drug-induced temporality of hashish, which enabled its users to experience an expanded present (Janet, 1928: 375). Paul Schilder argued that, lacking the immediate facts of 'Myself-Here-Now' (*Ich-Hier-Jetzt*), the presentist pathology of schizophrenia manifested an intact *Ich-Hier* but a disturbed *Jetzt-Hier* existence (Schilder, 1935: 265), paralleling Minkowski's suggestion that schizophrenics were unable to become in time because they were not orientated toward the future. Schizophrenia revealed a pathology of desynchronization – the inability to sense and to synthesize becoming with mathematical time and thus to resonate in harmony with the social world (Van Duppen, 2017: 390).

In contrast to depression, where patients experienced a malfunction of the time *experience*, schizophrenia was therefore defined by a malfunction of the time *concept*.

Patients disbelieved the reality of any form of time, whether internal *duréé* or clock-time, and were particularly prone to doubt the truth-value of time as it was observed in the clinic or hospital. This disturbance in the time concept was also reflected in compulsive feelings of death and rebirth, echoing the experience of recurrent or eternal temporalities earlier noted and related to the dissociation of patients from ambient becoming. Schizophrenic patients claimed that they died at certain times and were born again ('I feel I'm not continuous. I seem to be reborn every moment'; 'I died three times; I seemed to go right back'), but only 'certain people' noticed this breakage in time. Fischer concluded that these psychotic states left no trace ('*keine Spur*') of a 'past structure' in the mind, and were unable to reverse-allocate the present moment in another point of existence. They had no sense of an inner time-scale, whether of the external world or of their own consciousness (Fischer, 1929: 252). Patients constantly employed a language of dynamism – 'a little while ago, right away, lately, later' – and, due to the absence of any time-scale, proposed fantastic feats like the ability to travel to Argentina in five minutes or attend horse races at Longchamp before circling the world (Minkowski, 1926a: 561).

Paradoxically, however, disruptions in the timeline of *Ichzeit* and *Weltzeit* also carried a future aspect contrary to their absent past: many schizophrenics claimed to have powers of prophecy, which Israeli termed 'delusional utopianism' (Israeli, 1936: 21). This last group ranged in talents of divination, from those who believed they could foretell what people would say, to those who claimed to be able to predict when doors would open or catastrophic world events would occur. One woman claimed that she was 'the pivotal point of the new age', would marry the Prince of Wales, and become the head of the League of Nations, ushering in a new pacific period in world history (ibid.). A particularly gifted patient who maintained that he had metaphysical foresight declared he could 'prophesy anything that may happen within the next few years'. According to Lewis, this patient 'consider[ed] that the weeks this year [were] only as long as three days of former years; it [was] all controlled by a machine in the basement of the hospital which regulate[d] the physical universe', and he accordingly created a new calendar to deal with the situation (Lewis, 1932: 619). In fact, a disbelief in the reality of clock- or clinic-time and a belief in temporal recurrence prompted many schizophrenics to suggest alternative, personally derived time schema in its place: 'My head is a clock, an apparatus. I make the time, the new time, as it should be' (Israeli, 1936: 58; Schilder 1935: 266).

Although the temporality of schizophrenia as outlined by psychiatrists enacted a discursive attempt to pathologize the experience of time that it induced, schizophrenic descriptions of time and their questioning of the daily routines in clinics and hospitals nevertheless challenged the hegemony of clock-time. In often intellectually probing language, they disputed the absolutism of clock-time and queried the substance of time itself: 'Time slides into the past, the walls have tumbled down. . . . In the morning when I wake up, yes, how can I say it, the "disappearable" is there again; this torments me terribly. . . . But the "disappearable" of time is not there, and how can you take hold of time, when it was yesterday!' (Fischer, 1930). The 'breakage' of time for schizophrenics was arguably more extreme than for depressants, prompting psychoanalysts to define the disease as a 'disintegration' or 'disaggregation' of mental time (Bonaparte, 1939: 77–8). Schizophrenia was accordingly defined as a malfunction of the time concept that

assumed a form of malignant presentism, which (a) modified the domain of time in the brain; (b) prevented perception of duration or becoming; (c) obstructed the synthesis of *Ichzeit* and *Weltzeit*; and (d) prompted morbid rationalism or an obsession with quantitative timekeeping systems, whether internal or external (Israeli, 1936: 53).

Schizophrenic patients nonetheless exhibited a surprising sensitivity to the dislocation between their personal experience of time and the mode of clock-time reckoning observed around them. Unable to conceive of time as lived experience, schizophrenics were unable to maintain and synchronize the parallel time schemas of lived time and objective time, dwelling overwhelmingly in the latter. Ludwig Binswanger accordingly identified two speeds of 'being-in-the-world' for schizophrenics: a sense of present stasis and an abrupt sensory experience when certain thoughts and actions spurred them emotionally. Neither sensation was reconcilable with the experience of time flowing multidirectionally, as in *Ichzeit*, or linear-progressively as in *Weltzeit* (Binswanger, 1963; Hannibal, 1955: 607; Minkowski, 1970[1933]: 284). This paralysing split in temporal experience produced a feeling of stasis more intense than that exhibited in depression (Meerloo, 1935: 238). Most importantly of all, the temporality of schizophrenia indicated that pathologically altered ego states experienced different temporal aspects that Otto Hannibal called a 'crack in the boundary between self and the world' that 'broke' the experience of lived time. As a result of this loss of vital contact with reality, the temporality of schizophrenics produced a 'strong argument for the illusory character of time overall' (Hannibal, 1955: 610).

Coda: The temporality of other mental disorders

Depression and schizophrenia were merely two families of mental disease that fascinated psychologists. Other research foci were a curious mix of aphasia, dementia, and compulsive-obsessive disorders. Clinical psychologists like Minkowski and Janet argued that memory disorders were characterized by an erosion of past experience. Janet drew upon the theory of two psychologists to explain this deficiency: the 'local sign' theory of Wilhelm Wundt and the physiological theory of Louis Bard, who suggested memories had a 'time index' in addition to 'a local space index' (Bard, 1922). Janet theorized that patients suffering from aphasia were physiologically unable to locate mental 'date tags' for past experiences and consequently could not recall events in memory. Lacking 'local temporal signs', such patients were incapable of distinguishing temporal events in the mind and therefore lost a sense of their personal pasts (Janet, 1928: 352). Janet hazarded that this error in temporal deduction might be plausible evidence for the absence of local signs and therefore a physiological basis of time, or at least the complexity of a date-sign mnemonic system (ibid.: 24). When presented with stories that researchers considered to be part of the immediate future or present, some even placed the same narratives behind them, in the past, a mental feat Janet found particularly puzzling. One boy asserted that he had 'very accurate memories' of his mother's visit three days ago, when in reality his mother had never come to talk to him – in fact, he had never seen her (ibid.: 23–4). In general, aphasia and dementia were often defined as the opposite of schizophrenia. If in the latter patients retained a complete memory but experienced a disturbed 'present', dementia patients suffered from memory

disturbances but retained normal 'Myself-Here-Now' factors (Israeli, 1932: 486–7). The aphasic withdrew from everyday life and from the continuous flux of consciousness, while the dementia patent 'not only fail[ed] to distinguish himself from his psychic content, he also *fail[ed] to distinguish himself from the duration represented by this content*' (Courbon, 1927; Minkowski, 1970[1933]: 364).

Conversely, Minkowski noted that patients suffering from compulsive-obsessive or manic disorders had a 'marked tendency to situate [themselves] in time', noting that patient responses 'abound[ed] in expressions of a temporal order, such as *formerly, since, always, in two or three days, right now, five minutes ago, many times, yesterday, the day before yesterday*' (Minkowski 1970[1933]: 377–80). Indeed, compulsive patients tended to be obsessed with time in a much deeper sense. While schizophrenics were unable to conceive of time, compulsive patients thought about it constantly. One of Schilder's patients confessed to feeling time flowing so rapidly that sensations like *before* and *now* were almost meaningless, forcing himself to 'calculate . . . how short the time is' (Schilder, 1935: 263). Minkowski and Lewis suggested that manics suffered a durational disease that rendered them unable to process becoming or unfolding in time (Lewis, 1932: 616). Manics appeared to live only in the present. Yet this 'subduction' in mental life even cut out the present, for it too required a certain duration (Minkowski, 1970[1933]: 294). Hence a twenty-year-old female patient reminded herself continually that time flowed:

> Now, while I am talking with you, I think each time I say a word: 'past', 'past', 'past'. . . . The idea of my marriage is intolerable because I have to tell myself that the ceremony will last an hour. I can't understand how other people do things, connect them to precise points in time, and remain completely calm while doing it . . . when I see others moving, walking, for example; at each movement, at each step, I have to say, 'a second, another second'. (Von Gebsattel, 1928: 76)

Dutch psychologist Abraham Maurits Meerloo corroborated this relentless presentism, remarking that while neurotics were obsessed with clock-time, they tended to see world-time as a static block and exist without a past or future perspective (Meerloo, 1935: 240).

Mental disturbances beyond the confines of schizophrenia and depression were therefore characterized by competing temporalities. Although the associational network between past, present, and future was similarly disrupted, this disruption manifested itself in various forms. Aphasics and patients suffering from senile dementia, for example, were perceived to process mental life inappropriately. In an attempt to overcome this deficiency, some constructed a spurious past. Patients afflicted by compulsive or anxiety disorders, meanwhile, repeatedly emphasized their presentism in expressions of clock-time, yet failed to compute past experiences with the present. Uniting all case studies, however, was an engagement with time that questioned the standards of personal duration as an unceasing flow of past, present, and future, and of clock-time as the essential means by which to measure temporal experience. Although reaching diverse conclusions, these and other attempts to define mental illnesses on the basis of abnormal temporalities reveal how time had become a powerful analytical tool of interwar psychiatrists.

Conclusion

Partly inspired by Bergson and partly informed by a widespread engagement with time and temporality in the early 20th century, psychiatrists interested in mental diseases looked to phenomenology for new ways to understand human psychology. They believed that mental diseases had psychological as opposed to physiological causes; to understand the holism of human experience, psychiatry must necessarily pass through philosophy (Hönigswald, 1929: 715; Specht, 1912). Although they accepted Bergson's conception of duration as a dynamic psychological process, they adapted his philosophy to reflect existence in the social world. They asserted that time was given an inherent unity through life experience and that it therefore contained a 'stable' (clock-time or *Weltzeit*) as well as a 'dynamic' (ego-time or *Ichzeit*) aspect that could be empirically uncovered in psychopathologies, revealing the aetiology of mental diseases and the typical processes of normal mental life (Minkowski, 1970[1933]: xiv; Münsterberg, 1911). This spoke to Blondel's original assertion that the idea of time was a crucial example of how conscious life constructed a reality based on subjective experience, extracted from emotions or sensations that transformed an illusion of time or a timeline into a meaningful temporal universe (Blondel, 1914: 215).

Psychiatrists asserted that patients were unable to (a) locate their temporal context on a timeline; (b) estimate different portions of the timeline, particularly past or future; (c) perceive an appropriate 'static' time amidst flux; and/or (d) synthesize their own continuity of life within a universal time (Meerloo, 1935: 231–6). Unable to equilibrate internal (*Ichzeit*) and external (*Weltzeit*) time while conceiving of an open future and thus 'become' in time, the psychopathologic personality experienced an unnatural temporal world furnished by disturbances in relation to past, present, or future. These disruptions varied according to the particular disease – or rather, mental diseases were defined based on divergent disorientations toward the three dimensions of time. Phenomenological psychiatrists of the Minkowskian 'school' stressed the ability to anticipate an open future and to accommodate the flow of life as vital to a healthy psyche. By contrast, psychopathologies were characterized by a distortion in lived time and an inability to conceive of the future. This emphasis on the future shifted psychiatric thinking away from the prominence of the past in Freudian psychoanalysis or disturbances in the present that earlier French psychiatrists like Pierre Janet and Charles Blondel had proposed as the origins of mental disease, and focused on impediments to temporal becoming. Although Minkowski and like-minded psychiatrists asserted that pathologies developed from faulty psychologies, materialist psychologists like Fischer and Meerloo concluded that disturbances in the 'temporal field' were produced anatomically in the brain and linked to different functions that, working together, normally produced a 'localization in time' and 'consciousness of time' in healthy individuals (Meerloo, 1935). Which brings us back to the question: why was temporality an important locus of psychological debate?

Research on time pathologies represented more than a simple attempt to define or understand mental neuroses. Although most researchers failed to suggest ways of treating patients who suffered from psychopathologies of time (which presumably might, following this logic, have been achieved by rehabilitating them to clock-time), all agreed that studying the psychopathology of time would elucidate the 'ordinary' time-sense

central to human experience, and even predict rates of patient recovery (Israeli, 1936: 118). In this vein, two American psychologists tested each other in a soundproof room over periods of 48 and 86 hours to see if each could accurately record the time of day without external cues or timekeeping devices in solitary confinement. Under constant observation and lacking the ability to pass the time in a meaningful way (they were allowed to eat, sleep, and exercise, but not to read or to perform any other activity that 'might interfere with an even, uneventful existence'), Robert MacLeod and Merrill Roff concluded that though they both initially experienced an immense sense of temporal uncertainty, they came to develop their own personal time schemas that, although they were arbitrary, each adhered to 'without . . . confusion' (MacLeod and Roff, 1936: 396). With surprise, they noted how the absence of clock-time had not produced 'temporal vacuum', but prompted a new form of personal time-orientation – ironically failing to notice the structural similarities between the contexts of clinical confinement and the artificial isolation of their time laboratory, as well as how the suspension of quotidian occupations might alter one's time-sense (ibid.: 405).[2] Nonetheless, their experiment reinforced the theory that man possessed an internal and an external time-sense: though at the mercy of their own time consciousness, both MacLeod and Roff constructed arbitrary time systems to derive meaning from their predicament.

Understanding maladies in temporal experience was therefore a means by which researchers sought to understand and rescue their patients, an almost redemptive attempt to put them back on the path to self-realization (Pichon, 1931). As we have seen, Bergson's inception of psychological time in the form of *élan vital* carried metaphysical overtones. Knowledge of time was a critical means of constructing the self. As one patient noted, the flux and continuity of life 'clung' to everyone, allowing each man to 'find himself – isn't it true – in his own self', a mode of becoming that was impossible for schizophrenics, who felt themselves 'cut off' from the flow of time (Minkowski, 1970[1933]: 287–8). Moreover, patients were forced to insidiously operate in a linguistic world where they employed the same time-words and concepts as normal individuals, producing even more violent internal contradictions in their perception of the world and their ability to describe temporal experience (Blondel, 1914: 225). The divergence between personal time and clock-time testified by their patients, moreover, prompted some researchers to contemplate whether all temporal distinctions were artificial. Erwin Straus, for example, concluded that there was only ever a single time that was conceived or articulated in different ways: even clock-time was 'subjective' insofar as it was intersubjective and immanent (Moskalewicz, 2018: 5).

The psychopathological theories articulated by these psychiatrists, and, in particular, their emphasis on conceptions of the future as constitutive of ambient becoming, illustrate how temporality was an influential constituent of human experience in this period, laying the ground for future work on time-orientation, personality types, and child development (Campbell, 1934; Jaensch and Kretz, 1932; Murphy and Jensen, 1932; Piaget, 1946; Stern, 1930). Time pathologies also indicate the epistemological shift from experimental to pathological and phenomenological methods for understanding the human psyche in the history of psychology. Even though we may have lost some of this sense, contemporaries saw temporal malfunctions as better explanations for certain psychopathologies than, say, childhood trauma or repressed memories. At a moment

when psychologists are returning to the function of time in psychoses, and to Minkowski's work in particular (Abe and Raballo, 2013; Fuchs and Van Duppen, 2017), the history of the interaction between phenomenology, psychology, and psychiatry may present useful data for current practitioners seeking to understand methodological assumptions or weaknesses of earlier investigations, as well as the limitations of DSM-centred empiricism (Malhi and Kuiper, 2013; Popov and Popova, 2015; Stranghellini *et al.*, 2016; Thönes and Oberfeld, 2015). Psychopathologies of time reveal the centrality of time to early 20th-century thinking and the importance of phenomenology to psychiatry, further reminding us that attending to time can tell us something about the world and others who surround us.

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1. In Germany, Geentert Bouman and Adolf Grünbaum argued that the time of everyday life was characterized by a 'temporal complexus' with two aspects: one 'prospective', the other 'retrospective' (Bouman and Grünbaum, 1929).
2. John J. Harton, a psychologist at Arizona State Teachers College, conducted several experiments on the time-estimation of short and long durations among college students, concluding that compared to 'restless' or 'disturbed' periods of estimation, a 'unity' or homogeneous continuity of activity was associated with greater accuracy in estimating durations (Harton, 1939).

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